

Graduate Seminar 2

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Part 1 - Solving $\psi(x, t)$

- To this end, we need to resolve $\psi(x, t)$ which is given by the regular TDSE, i.e

$$i \frac{\partial}{\partial t} \psi(x, t) = \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \psi(x, t), \quad \psi(x, 0) = \psi_0(x), \quad (1)$$

$$\hat{H}_a = -\frac{1}{2} \frac{d^2}{dx^2} - \kappa \delta(x), \quad \hat{V}_{\text{ext}} = F(t)x, \quad (2)$$

$$E_0 = -\kappa^2/2, \quad \psi_0(x) = \kappa^{1/2} e^{-\kappa|x|}, \quad \kappa = 1 \quad (3)$$

$$E_k = k^2/2, \quad \psi_k^{(\pm)}(x) = e^{ikx} + R(\pm|k|)e^{\pm i|kx|}, \quad R(k) = \frac{i\kappa}{k - i\kappa}. \quad (4)$$

- The parameters of the laser we will be using are given by the relations

$$F(t) = F_0 e^{-(2t'/\tau)^2} \cos(\omega_0 t'), \quad 0 \leq t \leq T, \quad t' = t - T/2, \quad (5)$$

$$\tau = 2\pi n_{oc}/\omega_0, \quad n_{oc} = 5. \quad (6)$$

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Survival probability and PEMD

- The coefficients of the projection of $\bar{\Psi}$ are then, at T , and by definition, expressed as

$$a_0 = e^{iE_0T} \int \psi_0(x) \psi(x, T) dx, \quad a_k = e^{iE_kT} \int \psi_k^{(-)*}(x) \psi(x, T) dx. \quad (7)$$

- The survival probability is then expressed, at the end of the interaction, by

$$P_0 = |a_0|^2. \quad (8)$$

- The ionization probability can then be found many different ways that should be consistent with each others

$$P_{\text{ion}} = \int P(k) \frac{dk}{2\pi}, \quad \text{with} \quad P(k) = |a_k|^2. \quad (9)$$

$$= 1 - P_0. \quad (10)$$

- $P(k)$ is the probability for the electron being ionized with the energy E_k .

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Quantum spectrum of HHG photons (semi-classical approx.) - 1

- In the classical approx., the spectrum is obtained from the following (Eq.(80))

$$Q_{\nu}^{(c)}(\omega) = \alpha^2 \omega^2 |\mathbf{A}_{\nu} \mathbf{d}(\omega)|^2, \quad (11)$$

where for an arbitrary operator $\hat{f}$, we have

$$f(t) = \int \psi^*(x, t) \hat{f} \psi(x, t) dx \quad (12)$$

and the Fourier transform is obtained using

$$f(\omega) = \int f(t) e^{i\omega t} dt. \quad (13)$$

- Also, we consider the relations

$$\hat{p} = -id/dx, \quad \hat{d} = -x \quad (14)$$

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Quantum spectrum of HHG photons (semi-classical approx.) - 2

- Let us apply the following simplifications

$$\nu \longrightarrow \omega, \quad \mathbf{A}_\nu = \sqrt{\frac{2\pi c^2}{\omega L^3}} \mathbf{e}_\nu \longrightarrow A_\omega, \quad \text{and} \quad \alpha A_\omega = 1. \quad (15)$$

- So the observable we are considering here is

$$Q^{(c)}(\omega) = \omega^2 |d(\omega)|^2. \quad (16)$$

- Additionally let us consider Eq. (104)

$$Q^{(c)}(\omega) = |p(\omega)|^2. \quad (17)$$

and, by virtue of $p(t) = -\dot{d}(t)$, which reads

$$Q^{(c)}(\omega) = \left| -\dot{d}(\omega) \right|^2. \quad (18)$$

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Length, velocity and acceleration gauges

- (12) applied to the operator $\hat{d}$ gives its expression in the length gauge

$$d(t) = \langle \hat{d} \rangle = - \langle \hat{x} \rangle = - \int \psi^*(x, t) \hat{x} \psi(x, t) dx \quad (19)$$

- $\dot{d}(t)$ is given by the formula

$$\dot{d}(t) = \frac{d}{dt} d(t) \quad (20)$$

- $p(t)$ is given by the formula

$$p(t) = \langle \hat{p} \rangle = -i \int \psi^*(x, t) \frac{\partial}{\partial x} \psi(x, t) dx \quad (21)$$

- Using the Ehrenfest theorem for the operator x , we get the relations

$$\frac{d}{dt} \langle \hat{x} \rangle = \frac{\langle \hat{p} \rangle}{m}, \quad (22)$$

$$\frac{d}{dt} \langle \hat{p} \rangle = \left\langle -\frac{\partial V_{\text{ext}}}{\partial x} \right\rangle \quad (23)$$

and some others (see Bransden and Joachain p.531-532)

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Additional formula from Ehrenfest theorem

- Using the integration by parts formula, we get

$$p(\omega) = \int_{-T/2}^{T/2} e^{i\omega t} \frac{dx(t)}{dt} dt, \quad u(t) = x(t), \quad v(t) = e^{i\omega t}, \quad (24)$$

thus

$$p(\omega) = \int_{-T/2}^{T/2} \frac{d}{dt} \left(e^{i\omega t} x(t) \right) dt - i\omega \int_{-T/2}^{T/2} e^{i\omega t} x(t) dt, \quad (25)$$

$$= \left[e^{i\omega t} x(t) \right]_{-T/2}^{T/2} - i\omega \int_{-T/2}^{T/2} e^{i\omega t} x(t) dt. \quad (26)$$

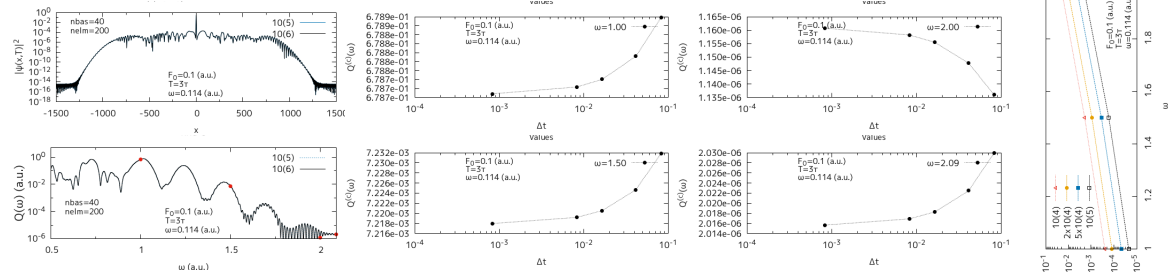
So, we get an additional formula for evaluating

$$Q^{(c)}(\omega) = |p(\omega)|^2, \quad (27)$$

with $p(\omega)$ evaluated conformly to Eq. (26).

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Time dependence - $T = 3\tau$

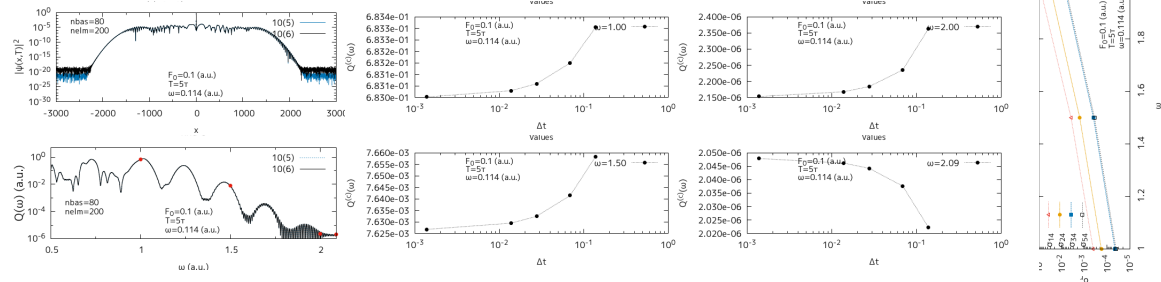


nsteps	10^4		2.10^4		5.10^4		10^5		10^6	
ω	$Q^{(c)}(\omega)[1]$	σ_{15}	$Q^{(c)}(\omega)[2]$	σ_{25}	$Q^{(c)}(\omega)[3]$	σ_{35}	$Q^{(c)}(\omega)[4]$	σ_{45}	$Q^{(c)}(\omega)[5]$	σ_{55}
1.00	0.6788784	0.0002509	0.6787917	0.0001231	0.6787404	0.0000475	0.6787234	0.0000225	0.6787081	-
1.50	7.2318868(-3)	0.0019271	7.2246750(-3)	0.000928	7.2205426(-3)	0.0003554	7.2191877(-3)	0.0001677	7.2179770(-3)	-
2.00	1.1359938(-6)	0.0213401	1.1477297(-6)	0.0112297	1.1554826(-6)	0.0045506	1.1582244(-6)	0.0021885	1.1607647(-6)	-
2.09	2.0299845(-6)	0.0071059	2.0224850(-6)	0.0033853	2.0182886(-6)	0.0013034	2.0169040(-6)	0.0006165	2.0156614(-6)	-

$$\sigma_{ij} = \left| \frac{Q^{(c)}(\omega)[i] - Q^{(c)}(\omega)[j]}{Q^{(c)}(\omega)[j]} \right|, \quad Q^{(c)}(\omega) = \left| \left[e^{i\omega t} x(t) \right]_{-T/2}^{T/2} - i\omega \int_{-T/2}^{T/2} e^{i\omega t} x(t) dt \right|^2.$$

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Time dependence - $T = 5\tau$

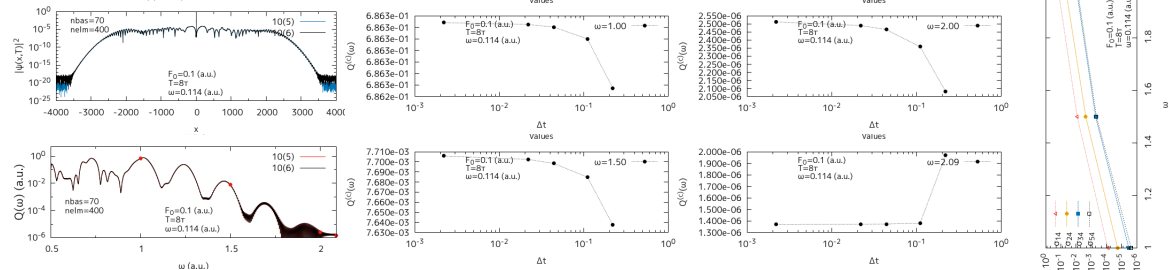


nsteps	10^4		$2 \cdot 10^4$		$5 \cdot 10^4$		10^5		10^6	
ω	$Q^{(c)}(\omega)[1]$	σ_{14}	$Q^{(c)}(\omega)[2]$	σ_{24}	$Q^{(c)}(\omega)[3]$	σ_{34}	$Q^{(c)}(\omega)[4]$	σ_{44}	$Q^{(c)}(\omega)[5]$	σ_{54}
1.00	0.683303	0.0004015	0.6831498	0.0001762	0.6830596	0.0000442	0.6830295	-	0.6830023	0.0000398
1.50	7.658297(-3)	0.0037774	7.6414614(-3)	0.0015707	7.6324580(-3)	0.0003906	7.6294776(-3)	-	7.6267840(-3)	-0.0003531
2.00	2.362065(-6)	0.0896783	2.2352490(-6)	0.0311749	2.1839406(-6)	0.0075051	2.1676721(-6)	-	2.1530603(-6)	-0.0067408
2.09	2.022331(-6)	-0.0710305	2.0376200(-6)	-0.0635588	2.0442287(-6)	-0.060329	2.0461706(-6)	-	2.0478578(-6)	0.0008246

$$\sigma_{ij} = \left| \frac{Q^{(c)}(\omega)[i] - Q^{(c)}(\omega)[j]}{Q^{(c)}(\omega)[j]} \right|, \quad Q^{(c)}(\omega) = \left| \left[e^{i\omega t} x(t) \right]_{-T/2}^{T/2} - i\omega \int_{-T/2}^{T/2} e^{i\omega t} x(t) dt \right|^2.$$

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Time dependence - $T = 8\tau$



nsteps	10^4		$2 \cdot 10^4$		$5 \cdot 10^4$		10^5		10^6	
ω	$Q^{(c)}(\omega)[1]$	σ_{14}	$Q^{(c)}(\omega)[2]$	σ_{24}	$Q^{(c)}(\omega)[3]$	σ_{34}	$Q^{(c)}(\omega)[4]$	σ_{44}	$Q^{(c)}(\omega)[5]$	σ_{54}
1.00	0.6862472	-0.0000803	0.6862899	-0.0000182	0.6863001	-0.0000033	0.6863024	-	0.6863040	0.0000023
1.50	7.6376879(-3)	-0.0083983	7.6849142(-3)	-0.0022669	7.6984587(-3)	-0.0005084	7.7023749(-3)	-	7.7058210(-3)	-0.0004472
2.00	2.0810428(-6)	-0.1641447	2.3582695(-6)	-0.0527959	2.4642315(-6)	-0.0102361	2.4897164(-6)	-	2.5118399(-6)	0.0088859
2.09	1.9677690(-6)	0.4343673	1.3812389(-6)	0.0068275	1.3729975(-6)	0.0008201	1.3718724(-6)	-	1.3710046(-6)	-0.0006326

$$\sigma_{ij} = \left| \frac{Q^{(c)}(\omega)[i] - Q^{(c)}(\omega)[j]}{Q^{(c)}(\omega)[j]} \right|, \quad Q^{(c)}(\omega) = \left| \left[e^{i\omega t} x(t) \right]_{-T/2}^{T/2} - i\omega \int_{-T/2}^{T/2} e^{i\omega t} x(t) dt \right|^2.$$

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Time dependence reference table

- Sufficient time step for converged result is $\Delta t \leq 0.08$ a.u.
- In the following, we will maintain Δt set to around 0.04 a.u.

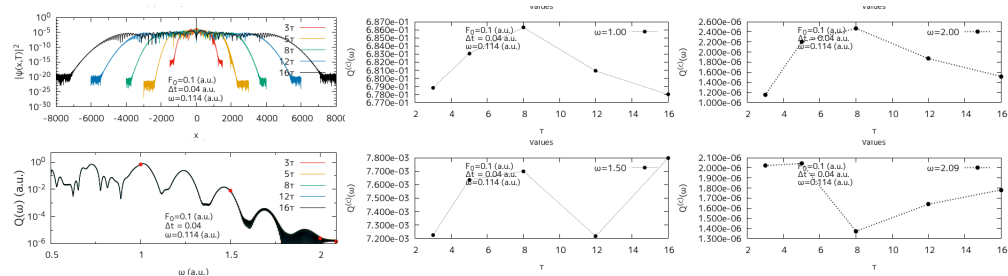
$\Delta t = 0.04$ a.u. $\omega = 0.057$ a.u. $\tau = 551.15661$									
$n = 3$		$n = 5$		$n = 8$		$n = 12$		$n = 16$	
$T = 3\tau$	nsteps	$T = 5\tau$	nsteps	$T = 8\tau$	nsteps	$T = 12\tau$	nsteps	$T = 16\tau$	nsteps
1653.4698	41336.745	2755.783	68894.576	4409.2528	110231.32	6613.8793	165346.98	8818.5057	220462.64

$\Delta t = 0.04$ a.u. $\omega = 0.114$ a.u. $\tau = 275.5783$									
$n = 3$		$n = 5$		$n = 8$		$n = 12$		$n = 16$	
$T = 3\tau$	nsteps	$T = 5\tau$	nsteps	$T = 8\tau$	nsteps	$T = 12\tau$	nsteps	$T = 16\tau$	nsteps
826.73491	20668.373	1377.8915	34447.288	2204.6264	55115.661	3306.9396	82673.491	4409.2528	110231.32

$$\tau = n_{oc} \frac{2\pi}{\omega}, \quad T = n\tau, \quad \Delta t = \frac{T}{\text{nsteps}}, \quad n_{oc} = 5,$$

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Time dependence - $F_0 = 0.1$ a.u. $\Delta t = 0.04$ a.u.



$F_0 = 0.1$ a.u. $\Delta t = 0.04$ a.u., $\tau = 5 \times 2\pi/\omega = 275.5783$ a.u.						
n	$T = n\tau$	nsteps	$Q^{(c)}(\omega = 1.00)$	$Q^{(c)}(\omega = 1.50)$	$Q^{(c)}(\omega = 2.00)$	$Q^{(c)}(\omega = 2.09)$
3	826.73491	20668	0.678788979588210	7.224449588416915(-3)	1.148130921051377(-6)	2.022256187878468(-6)
5	1377.8915	34448	0.683086925522419	7.635146846170220(-3)	2.198806116374674(-6)	2.042395189629913(-6)
8	2204.288	55116	0.686300611013558	7.699195832628777(-3)	2.469101583853556(-6)	1.372771210289770(-6)
12	3306.9396	82674	0.680910244964746	7.217558844953091(-3)	1.868727433549037(-6)	1.639897289171694(-6)
16	4409.2528	110232	0.677994169556346	7.799224872958270(-3)	1.517317908696027(-6)	1.781572497404902(-6)

$$Q^{(c)}(\omega) = \left| \left[e^{i\omega t} x(t) \right]_{-T/2}^{T/2} - i\omega \int_{-T/2}^{T/2} e^{i\omega t} x(t) dt \right|^2$$

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Time dependence - $F_0 = 0.3 \text{ a.u.}$ $\Delta t = 0.04 \text{ a.u.}$

$F_0 = 0.3 \text{ a.u.}$ $\Delta t = 0.04 \text{ a.u.},$ $\tau = 5 \times 2\pi/\omega = 275.5783 \text{ a.u.}$						
n	$T = n\tau$	nsteps	$Q^{(c)}(\omega = 1.00)$	$Q^{(c)}(\omega = 1.50)$	$Q^{(c)}(\omega = 2.00)$	$Q^{(c)}(\omega = 2.09)$
3	826.73491	20668				
5	1377.8915	34448				
8	2204.288	55116				
12	3306.9396	82674				
16	4409.2528	110232				

$$Q^{(c)}(\omega) = \left| \left[e^{i\omega t} x(t) \right]_{-T/2}^{T/2} - i\omega \int_{-T/2}^{T/2} e^{i\omega t} x(t) dt \right|^2.$$

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Convergence parameters table for the homogeneous equation

- The following table shows parameters known to return converged results for Eq. (1), which is recalled below

$\omega = 0.057 \text{ a.u.}, \quad \Delta t \leq 0.08 \text{ a.u.}, \quad \tau = 5 \times 2\pi/\omega = 551.15661 \text{ a.u.}$					
$F_0(\text{a.u.})$	3τ	5τ	8τ	12τ	16τ
0.03	(20/200) - 3000	(60/200) - 7000	(80/200) - 10000	(90/200) - 14000*	(90-300) - 16000*
0.1	(70/400) - 8000	(100/500) - 14000	?		
0.3	?				
1.0					

Legend

(nbas/nelm) – xmax

$\omega = 0.114 \text{ a.u.}, \quad \Delta t \leq 0.08 \text{ a.u.}, \quad \tau = 5 \times 2\pi/\omega = 275.5783 \text{ a.u.}$					
$F_0(\text{a.u.})$	3τ	5τ	8τ	12τ	16τ
0.03	(20/100) - 2000	(40/200) - 4000	(40/200) - 6000	(80/200) - 8000	(70-200) - 10000
0.1	(40/200) - 3000	(80/200) - 6000	(70/300) - 8000	(90/400) - 12000	(70-400) - 12000*
0.3	(90/400) - 7000	(100/500) - 10000			
1.0					

$$i \frac{\partial}{\partial t} \psi(x, t) = \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \psi(x, t), \quad \psi(x, 0) = \psi_0(x),$$

$$\hat{H}_a = -\frac{1}{2} \frac{d^2}{dx^2} - \kappa \delta(x), \quad \hat{V}_{\text{ext}} = F(t)x,$$

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Structure of the code, performance and memory management

1. Diagonalization

- most time consuming part for bigger matrixes
- is there a way to multithread lapack diagonalization?

2. Propagation

- can be arbitrarily long (convergence requires $\Delta t \leq 0.08$ a.u.)
- order 4 scheme required
- multithreading would also help here

3. Projection

- very short (matter of minutes)
- potentially very memory consuming

1. and 2. take some memory for a long time, 3. takes even more memory for a very short time.
Any run for large matrixes required to split those processes.

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Numerical toolkit - 1

■ Integration is done by using

- the (1/3) Simpson rule, m being the number of subintervals with $h = (x_m - x_0)/m$,
 $x_i = x_0 + ih$,

$$\int_{x_0}^{x_m} f(x)dx \sim \frac{1}{3}h \sum_{i=1}^{m/2} [f(x_{2i-2}) + 4f(x_{2i-1}) + f(x_{2i})] \quad (1/3 \text{ rule}) \quad (28)$$

- the trapezoidal rule

$$\int_{x_0}^{x_m} f(x)dx \sim \sum_{i=1}^m \frac{f(x_{i-1}) + f(x_i)}{2} \Delta x_i \quad (29)$$

■ Both those rules are of order 2

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Numerical toolkit - 2

■ Finite differences

$$\begin{cases} f'_l(x_i) = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}, & f'_r(x_i) = \frac{f(x_i) - f(x_{i-1}))}{x_i - x_{i-1}} \\ f'(x_i) = \frac{f'_l(x_i) + f'_r(x_i)}{2} \end{cases} \quad (30)$$

■ Using the reciprocal space for differentiating (Fourier transform)

$$\hat{f}(k, t) = \int_{-\infty}^{\infty} f(x, t) e^{-ikx} dx, \quad \Rightarrow \quad \frac{\partial}{\partial x} f(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} ik \hat{f}(k, t) e^{ikx} dk, \quad (31)$$

■ Let us also write the dispersion relation

$$\omega = \frac{k^2}{2} \quad \Rightarrow \quad k = \pm \sqrt{2\omega} \quad (32)$$

■ The delta function is evaluated as

$$\delta(k - k') = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{i(k-k')x} dx \quad (33)$$

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Part 2 - Solving $\phi_\omega(x, t)$

- We are now interested in solving the following 1D problem with a ZRP

$$\begin{cases} i \frac{\partial}{\partial t} \psi(x, t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \psi(x, t), \\ i \frac{\partial}{\partial t} \phi_\omega(x, t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \phi_\omega(x, t) + e^{i\omega t} \hat{p} \psi(x, t), \end{cases} \quad (34)$$

where

$$\hat{H}_a = -\frac{1}{2} \frac{d^2}{dx^2} - \kappa \delta(x), \quad \hat{V}_{\text{ext}}(t) = F(t)x, \quad (35)$$

$$\phi_\omega(x, 0) = \frac{-i\kappa^{3/2}}{\omega} \text{sgn}(x) \left(e^{-\kappa|x|} - e^{-\kappa(\omega)|x|} \right), \quad \psi(x, 0) = \kappa^{1/2} e^{-\kappa|x|}, \quad (36)$$

with

$$\hat{p} = -id/dx, \quad \text{and} \quad \kappa(\omega) = \sqrt{\kappa^2 + 2\omega}. \quad (37)$$

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Solutions to the homogeneous and inhomogeneous equations

$$\begin{cases} i \frac{\partial}{\partial t} \psi(x, t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \psi(x, t), \\ i \frac{\partial}{\partial t} \phi_\omega(x, t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \phi_\omega(x, t) + e^{i\omega t} \hat{p} \psi(x, t), \end{cases}$$

$$\psi(x, t) = e^{-i \int_{t_0}^t (\hat{H}_a + \hat{V}_{\text{ext}}(\tau)) d\tau} \psi(x, 0), \quad (38)$$

$$\phi_\omega(x, t) = e^{-i \int_{t_0}^t (\hat{H}_a + \hat{V}_{\text{ext}}(\tau)) d\tau} \phi_\omega(x, 0) - i e^{-i \int_{t_0}^t (\hat{H}_a + \hat{V}_{\text{ext}}(\tau)) d\tau} \int_{t_0}^t e^{i \int_{t_0}^s (\hat{H}_a + \hat{V}_{\text{ext}}(\tau)) d\tau} e^{i\omega s} \hat{p} \psi(x, s) ds \quad (39)$$

$$\begin{aligned} \psi(x, t) &= U(t, t_0) \psi(x, 0), & U(t, t_0) &= e^{-i \int_{t_0}^t (\hat{H}_a + \hat{V}_{\text{ext}}(\tau)) d\tau} \\ \phi_\omega(x, t) &= U(t, t_0) \phi_\omega(x, 0) - U(t, t_0) \int_{t_0}^t U(s, t_0)^{-1} e^{i\omega s} \frac{\partial}{\partial x} \psi(x, s) ds \end{aligned} \quad (40)$$

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Time propagation

$$\begin{cases} i \frac{\partial}{\partial t} \psi(x, t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \psi(x, t), \\ i \frac{\partial}{\partial t} \phi_\omega(x, t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \phi_\omega(x, t) + e^{i\omega t} \hat{p} \psi(x, t), \end{cases}$$

$$\psi(x, t + \Delta t) = U(t + \Delta t, t) \psi(x, t),$$

$$\phi_\omega(x, t + \Delta t) = U(t + \Delta t, t) \phi_\omega(x, t) - \int_t^{t+\Delta t} U(t + \Delta t, s) e^{i\omega s} \frac{\partial}{\partial x} \psi(x, s) ds$$

■ From the trapezoidal rule applied to the integral, we get

$$\psi(x, t + \Delta t) = U(\Delta t) \psi(x, t), \quad U(t + \Delta t, t) = U(\Delta t) \quad (41)$$

$$\phi_\omega(x, t + \Delta t) = U(\Delta t) \phi_\omega(x, t) - \frac{\Delta t}{2} \left(e^{i\omega(t+\Delta t)} \frac{\partial}{\partial x} \psi(x, t + \Delta t) + U(\Delta t) e^{i\omega t} \frac{\partial}{\partial x} \psi(x, t) \right) \quad (42)$$

$$= U(\Delta t) \left(\phi_\omega(x, t) - \frac{\Delta t}{2} e^{i\omega t} \frac{\partial}{\partial x} \psi(x, t) \right) - \frac{\Delta t}{2} e^{i\omega(t+\Delta t)} \frac{\partial}{\partial x} \psi(x, t + \Delta t) \quad (43)$$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Projections

$$p_{k0} = \int \psi_k^{(-)*}(x) \hat{p} \psi_0(x) dx = \frac{2k\mathcal{K}^{3/2}}{\mathcal{K}^2 + k^2} \quad (44)$$

$$p_{kk'} = \int \psi_k^{(-)*}(x) \hat{p} \psi_{k'}^{(-)}(x) dx = 2\pi k \delta(k - k') - \frac{2\mathcal{K}}{k^2 - k'^2 + i0} \left(\frac{k'|k|}{|k| - i\mathcal{K}} - \frac{k|k'|}{|k'| + i\mathcal{K}} \right) \quad (45)$$

$$b_{0\omega}(T) = e^{iE_0 T} \int \psi_0(x) \phi_\omega(x, T) dx, \quad (46)$$

$$b_{k\omega}(T) = e^{iE_k T} \int \psi_k^{(-)*}(x) \phi_\omega(x, T) dx, \quad (47)$$

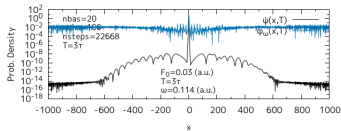
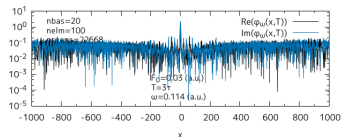
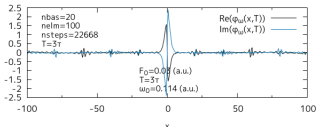
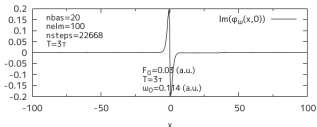
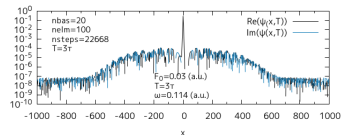
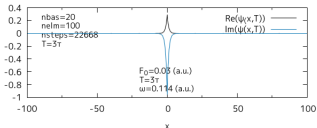
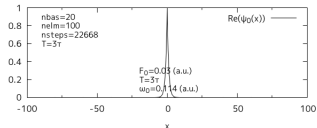
$$b_{0\omega} = b_{0\omega}(T) + \int \frac{e^{i(E_0 + \omega - E_{k'})T} p_{0k'} a_{k'}}{E_0 + \omega - E_{k'} + i0} \frac{dk'}{2\pi} \quad (48)$$

$$b_{k\omega} = b_{k\omega}(T) + \frac{e^{i(E_k + \omega - E_0)T} p_{k0} a_0}{E_k + \omega - E_0} + \int \frac{e^{i(E_k + \omega - E_{k'})T} p_{kk'} a_{k'}}{E_k + \omega - E_{k'} + i0} \frac{dk'}{2\pi} \quad (49)$$

$$Q(\omega) = |b_{0\omega}|^2 + \int |b_{k\omega}|^2 \frac{dk}{2\pi} \quad (50)$$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

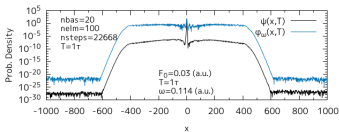
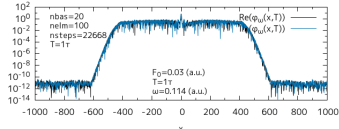
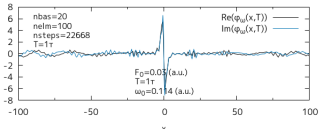
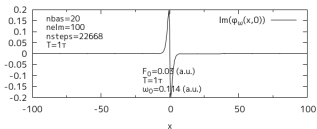
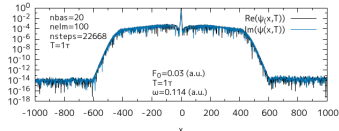
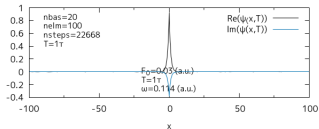
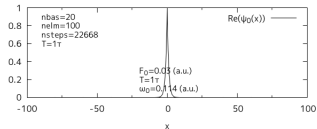
The wavefunction $\phi_\omega(x, t)$ ($\omega = 1$)



nbas = 20
nelm = 100
nsteps = 22668
xmax = 2000
T = 3τ

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

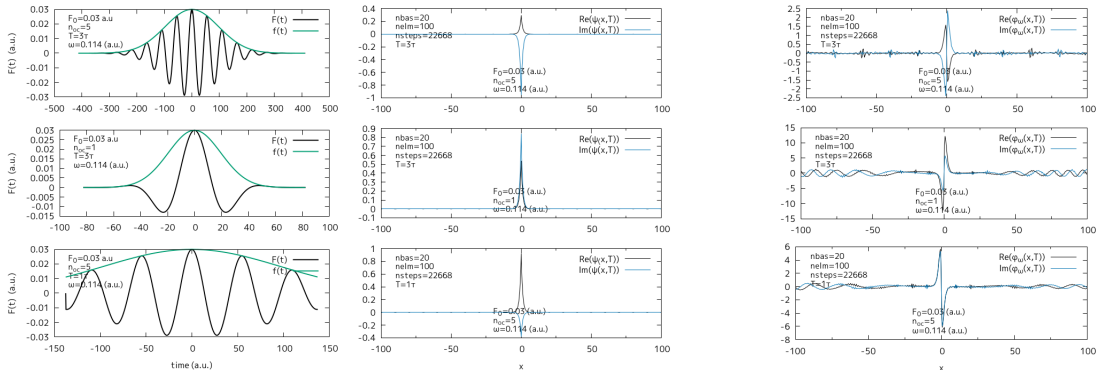
The wavefunction $\phi_\omega(x, t)$ ($\omega = 1$)



nbas = 20
nelm = 100
nsteps = 22668
xmax = 2000
T = 1τ

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

About the field



Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

The FEDVR basis - 1 - basic properties

- From Ohmi *et al*, Phys. Rev. A 92 (2015), we get

$$\pi_i^s(r) = \frac{1}{\sqrt{\omega_i}} \prod_{\substack{k=1 \\ k \neq i}}^{N_p} \frac{r - r_k^s}{r_i^s - r_k^s}, \quad \pi_i^s(r_k^s) = \delta_{ik} / \sqrt{\omega_i^s}, \quad (51)$$

$$\left. \frac{d\pi_i^s(r)}{dr} \right|_{r=r_j^s} = \frac{1}{\sqrt{\omega_i}} \begin{cases} \frac{1}{r_i^s - r_j^s} \prod_{\substack{k=1 \\ k \neq i, j}}^{N_p} \frac{r_j^s - r_k^s}{r_i^s - r_k^s}, & i \neq j, \\ \frac{1}{2} \left(\left[\pi_i^s(r_{N_p}^s) \right]^2 - \left[\pi_i^s(r_1^s) \right]^2 \right), & i = j, \end{cases} \quad (52)$$

- N_s sectors $r_0 \leq \dots r_s \leq r_{N_s}$ and N_p nodes a sector, which makes the number of point N so

$$N = N_s(N_p - 1) + 1 \quad (53)$$

$$n = (s - 1)(N_p - 1) + i \quad (54)$$

with

$$i = 1, \dots, N_p, \quad s = 1, \dots, N_s, \quad n = 1, \dots, N$$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

The FEDVR basis - 2 - The global basis

- The global FEDVR basis is defined, for $n = (s - 1)(N_p - 1) + N_p$, with $s = 1, \dots, N_s - 1$

$$\Pi_n(r) = \frac{1}{\sqrt{\Omega_n}} \begin{cases} \sqrt{\omega_{N_p}^s} \pi_{N_p}^s(r), & r \in [r_{s-1}, r_s], \\ \sqrt{\omega_1^{s+1}} \pi_1^{s+1}(r), & r \in [r_s, r_{s+1}], \end{cases} \quad \Omega_n = \omega_{N_p}^s + \omega_1^{s+1}, \quad (55)$$

This function bridges the sectors s and $s + 1$ and is continuous through the boundary between them (but its derivative is not).

- For the other values of n corresponding to the quadrature points inside sectors, i.e.

$$n = (s - 1)(N_p - 1) + i \quad (56)$$

and at the outer boundary $r_N = r_m$ of the box, we have

$$\Pi_n(r) = \pi_i^s(r), \quad r \in [r_s, r_{s+1}], \quad \Omega_n = \omega_i^s, \quad (57)$$

$$\Pi_n(r_{n'}) = \delta_{nn'} / \sqrt{\Omega_n}, \quad (58)$$

$$\int_0^{r_m} \Pi_n(r) f(r) \Pi_{n'}(r) dr \sim f(r_n) \delta_{nn'}. \quad (59)$$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

The FEDVR basis - 3 - the derivative

- We can extend Eq. (52) to any r of the domain and write

$$\frac{d\pi_i^s(r)}{dr} = \frac{1}{\sqrt{\omega_i^s}} \sum_{\substack{j=1 \\ j \neq i}}^{N_p} \frac{1}{r_i^s - r_j^s} \prod_{\substack{k=1 \\ k \neq j, i}}^{N_p} \frac{r - r_k^s}{r_i^s - r_k^s}, \quad (60)$$

- In terms of Π_n , the derivative is expressed as, for $n = (s-1)(N_p-1) + N_p$, with $\Omega_n = \omega_{N_p}^s + \omega_1^{s+1}$

$$\frac{d\Pi_n(r)}{dr} = \frac{1}{\sqrt{\Omega_n}} \begin{cases} \sqrt{\omega_{N_p}^s} d\pi_{N_p}^s(r)/dr, & r \leq r_s, \\ \sqrt{\omega_1^{s+1}} d\pi_1^{s+1}(r)/dr, & r \geq r_s, \end{cases} \quad (61)$$

and reduced to (60) otherwise.

The quantities we are looking for are

$$\psi(x, t) = \sum_{n=1}^N c_n(t) \Pi_n(x), \quad \text{and} \quad \frac{d}{dx} \psi(x, t) = \sum_{n=1}^N c_n(t) \frac{d}{dx} \Pi_n(x). \quad (62)$$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

The FEDVR basis - The two basis

$$n = (s - 1)(N_p - 1) + i$$

$$N = N_s(N_p - 1) + 1, \quad N' = N_s(N_p - 1) - 1$$

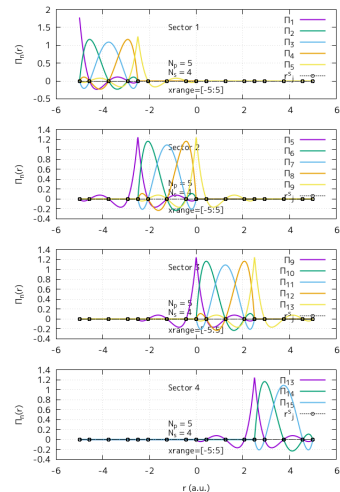
$$N_s = 4, \quad N_p = 5, \quad N = 17, \quad N' = 15.$$

- $\mathcal{B}$ provides mathematical support
- $\mathcal{B}'$ is where we have the expression of the wavefunction
- We note that $n' = n - 1$ when defined

$\mathcal{B}$				$\mathcal{B}'$			
n	s	i	x_n	n'	s'	i'	$x_{n'}$
1	1	1	-5.0000000000000000	1	1	1	-4.5683170883849709
2	1	2	-4.5683170883849709	2	1	2	-3.7500000000000000
3	1	3	-3.7500000000000000	3	1	3	-2.9316829116150291
4	1	4	-2.9316829116150291	4	1	4	-2.5000000000000000
5	1	5	-2.5000000000000000				
5	2	1	-2.5000000000000000	5	2	1	-2.0683170883849713
6	2	2	-2.0683170883849713	6	2	2	-1.2500000000000000
7	2	3	-1.2500000000000000	7	2	3	-0.43168291161502864
8	2	4	-0.43168291161502864	8	2	4	0.0000000000000000
9	2	5	0.0000000000000000				
9	3	1	0.0000000000000000	9	3	1	0.43168291161502864
10	3	2	0.43168291161502864	10	3	2	1.2500000000000000
11	3	3	1.2500000000000000	11	3	3	2.0683170883849713
12	3	4	2.0683170883849713	12	3	4	2.5000000000000000
13	3	5	2.5000000000000000				
13	4	1	2.5000000000000000	13	4	1	2.9316829116150291
14	4	2	2.9316829116150291	14	4	2	3.7500000000000000
15	4	3	3.7500000000000000	15	4	3	4.5683170883849709
16	4	4	4.5683170883849709				
17	4	5	5.0000000000000000				

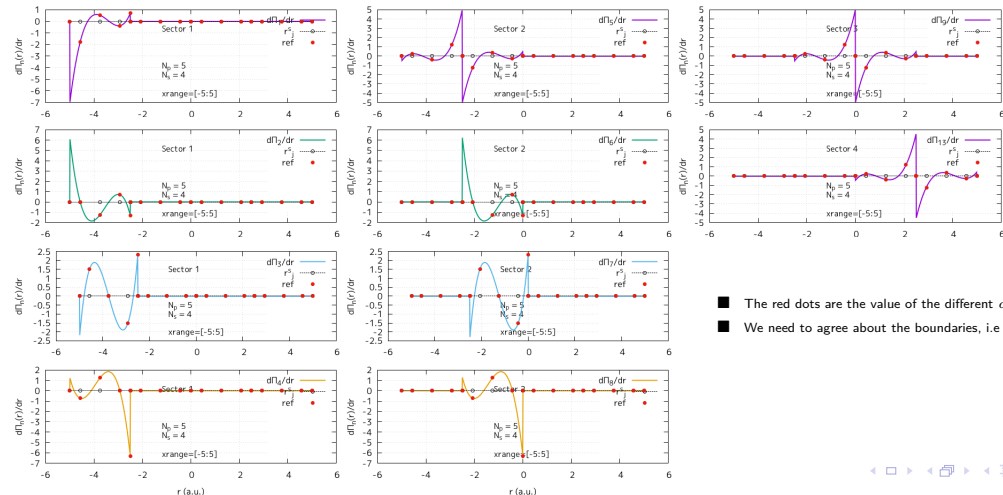
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The FEDVR basis - The $\mathcal{B}$ global basis



Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

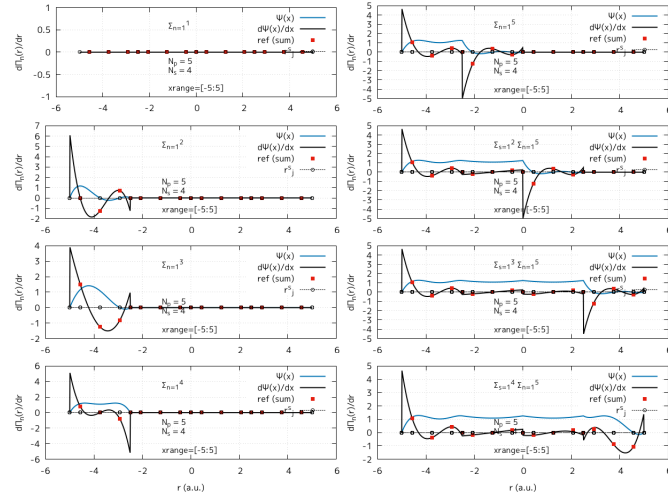
The FEDVR basis - The $\mathcal{B}$ global basis - the derivatives



- The red dots are the value of the different $d\Pi_n/dx$ around x_n
- We need to agree about the boundaries, i.e for $j = 1$ and $j = N_p$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

The FEDVR basis - The wavefunction on the $\mathcal{B}'$ basis (Construction)



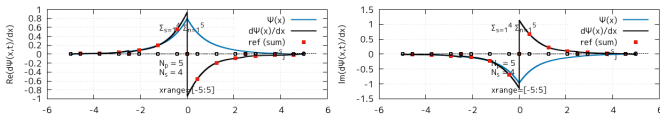
$$\psi(x, t) = \sum_{n'=1}^{N'} c_{n'}(t) \Pi_{n'}(x),$$

$$\frac{d}{dx} \psi(x, t) = \sum_{n'=1}^{N'} c_{n'}(t) \frac{d}{dx} \Pi_{n'}(x).$$

- We try to model (62)
- Let's say that all the $c_{n'}(t) = 1$
- $n' = n - 1$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

The FEDVR basis - The analytical wavefunction and derivative



$$\psi(x, t) = \sum_{n'=1}^{N'} c_{n'}(t) \Pi_{n'}(x),$$

$$\frac{d}{dx} \psi(x, t) = \sum_{n'=1}^{N'} c_{n'}(t) \frac{d}{dx} \Pi_{n'}(x).$$

$$\psi(x, t) = \text{Re} \{ \psi(x, t) \} + i \text{Im} \{ \psi(x, t) \}$$

- Rigorously (62) with non zero $c_n(t)$'s
- The bottom panel is the converged result

$$F_0 = 0.1 \text{ a.u.}$$

$$\text{nbas} = 20$$

$$\text{nelm} = 100$$

$$\text{nsteps} = 20668$$